Math 105L **Lab 7**

Inverse Functions & Graphical Symmetry

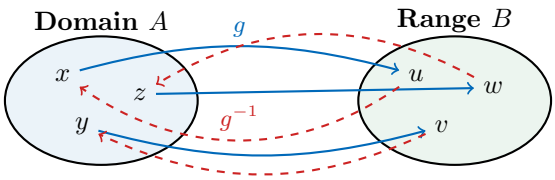
Part I: Invertible Functions & The Definition of an Inverse

- **Core Definition:** Suppose f is a function and there is a corresponding function g such that $g(f(t)) = t$ and $f(g(t)) = t$ for every value of t in their respective domains. Then g is called the **inverse function** of f, denoted $\mathbf{f^{-1}}$:

$$\mathbf{f\big(f^{-1}(x)\big) = x = f^{-1}\big(f(x)\big)}$$

Crucial Caution from Board: Note that $\mathbf{f^{-1}(x) \neq \frac{1}{f(x)}}$! The exponent -1 denotes functional inversion, not the multiplicative reciprocal.

Mapping Diagram Exploration: Suppose g is a function with domain $A = \{x, y, z\}$ and range $B = \{u, v, w\}$ such that $g(x) = u$, $g(y) = v$, and $g(z) = w$:



- **Describing g^{-1}:** Since $g^{-1}(g(t)) = t$, we must have:

$$g^{-1}(u) = x, \qquad g^{-1}(v) = y, \qquad g^{-1}(w) = z$$

- **Domain and Range Swap:**

$$\text{Domain}\big(g^{-1}\big) = \text{Range}(g) = \{u, v, w\}, \qquad \text{Range}\big(g^{-1}\big) = \text{Domain}(g) = \{x, y, z\}$$

- **Is g^{-1} a function?** Yes, because each input in $\{u, v, w\}$ points to exactly one output in $\{x, y, z\}$.

GeoGebra Exploration: Reflection Across $y = x$ & Strict Monotonicity

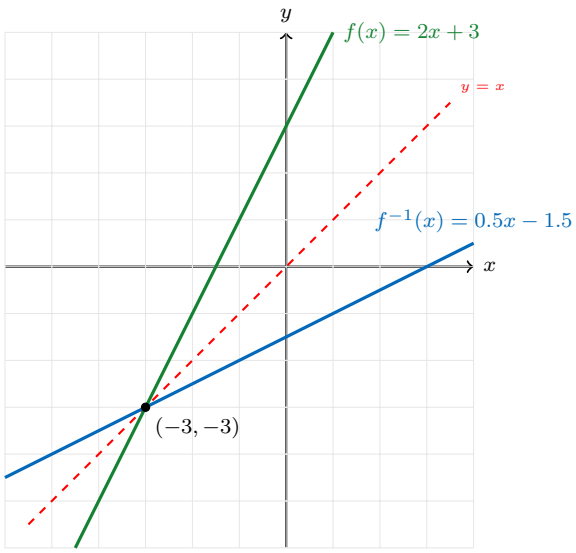
GeoGebra Investigation: Symmetry across the Line $y = x$

In the computer laboratory demonstration, we graph linear and non-linear inverse pairs:

Linear Pair Examined:

- $f(x) = 2x + 3$
- Solve for x: $y = 2x + 3 \implies 2x = y - 3 \implies x = \frac{1}{2}y - \frac{3}{2}$
- Inverse function: $\mathbf{g(x) = f^{-1}(x) = 0.5x - 1.5}$
- Line of symmetry: $\mathbf{h(x) = x}$ (red dashed diagonal).
- **Intersection Point:** The graphs of $f(x)$ and $f^{-1}(x)$ intersect on the line $y = x$ at:

$$2x + 3 = x \implies x = -3 \implies (\mathbf{-3, -3})$$



Invertibility Without an Explicit Radical Formula: $f(x) = x^5 + x$

Can a function be invertible even if we cannot algebraically isolate x?

- Consider the polynomial $f(x) = x^5 + x$.
- By the Abel-Ruffini theorem, there is no general formula in radicals to solve $x^5 + x = y$ for x.
- **Calculus Verification of Invertibility:**

$$f'(x) = 5x^4 + 1$$

Since $x^4 \geq 0$ for all real numbers x, we have $5x^4 \geq 0$, which guarantees:

$$\mathbf{f'(x) \geq 1 > 0} \quad \text{for all } \mathbf{x} \in (-\infty, \infty)$$

- Because the derivative is strictly positive everywhere, $f(x)$ is **strictly increasing** across its entire domain $(-\infty, \infty)$.
- Any strictly increasing function **must pass the Horizontal Line Test (HLT)**.
- **Conclusion:** The inverse function $f^{-1}(x)$ exists, is strictly increasing, and is continuous and differentiable everywhere, even without an elementary algebraic expression!

The Derivative Connection to Invertibility (Preparation for IFT):
If $f'(x) > 0$ everywhere (or $f'(x) < 0$ everywhere) on an interval, then $f(x)$ is strictly monotonic, passes the horizontal line test, and is guaranteed to be invertible on that interval.